
Lotus Vision Opticals CRM

Complete Features Documentation

Executive Summary

CRM Name: Lotus Vision Opticals

Tech Stack: Next.js 16, Turbopack, shadcn/ui, SQLite/Prisma ORM, Tailwind CSS

Total Codebase: ~20,855 lines across 16 files

Components: 14 CRM Components

API Routes: 21 API Routes

Database Models: 16 Prisma Models

Design Philosophy: Mobile-first responsive design with 44px touch targets

This document provides a comprehensive feature-level breakdown of every module in the Lotus Vision Opticals CRM system. The application is designed as a single-page application built on Next.js 16 with Turbopack for optimized development and build performance. It leverages shadcn/ui component library for consistent, accessible UI components styled with Tailwind CSS. Data persistence is handled through SQLite via Prisma ORM, providing a lightweight yet robust database solution suitable for optical retail operations.

Module-by-Module Features

1. Dashboard (1,235 lines)

Feature	Description	Status
KPI Summary Cards	Real-time display of total revenue, total sales, total customers, and active inventory count with period-over-period comparison indicators.	Implemented
Sales Trend Chart	Interactive line/area chart showing daily, weekly, and monthly sales trends with configurable date ranges.	Implemented
Top Products Widget	Ranked list of best-selling products by quantity and revenue within a selected time period.	Implemented
Low Stock Alerts	Visual warning indicators for products that have fallen below their configured minimum stock threshold.	Implemented
Recent Activity Feed	Chronological feed of the latest transactions, customer additions, and inventory changes with timestamps.	Implemented
Quick Action Shortcuts	One-click navigation buttons to create a new sale, add a customer, or manage inventory directly from the dashboard.	Implemented
Revenue Breakdown	Visual breakdown of revenue by payment mode (Cash, UPI, Card) and by product category.	Implemented

2. Customers (1,961 lines)

Feature	Description	Status
Full CRUD Operations	Create, read, update, and delete customer records with validation on all required fields including name, phone, and email.	Implemented
WhatsApp Integration	Direct WhatsApp message button on each customer card for instant communication using the customer's registered phone number.	Implemented
CSV Export with Date Filters	Export the full customer list to CSV format with optional date range filtering for targeted data extraction.	Implemented
Duplicate Phone Detection	Automatic detection and warning when a new or edited customer record contains a phone number that already exists in the database.	Implemented
Group Filter	Filter customers by group classification: Regular, Wholesale, New, enabling segment-specific views and management.	Implemented
Sortable Columns	All table columns (name, phone, group, date added) are sortable in ascending or descending order.	Implemented
Prescription Management	View and manage customer prescriptions directly from the customer detail view with spherical, cylindrical, and axis values.	Implemented
Visit History	Complete chronological log of all customer visits including purchase history, prescriptions issued, and appointments attended.	Implemented
Birthday Tracking	Record and track customer birthdays with upcoming birthday reminders for personalized outreach and campaign targeting.	Implemented
Search and Pagination	Full-text search across customer name and phone fields with configurable page size and navigation controls.	Implemented
Customer Detail View	Comprehensive single-customer view consolidating contact info, purchase history, prescriptions, and appointment records.	Implemented

3. Sales / POS (2,255 lines)

Feature	Description	Status
Invoice Creation	Full invoice creation workflow with product selection, quantity entry, price override, and automatic subtotal/tax/total calculation.	Implemented
Split Payment	Support for splitting a single invoice payment across multiple modes (e.g., part cash, part UPI) with individual amount tracking.	Implemented
Return Processing	Process sales returns with full or partial item returns, automatic refund calculation, and restocking of returned inventory.	Implemented
Invoice Print View	Dedicated print-optimized invoice layout suitable for A4 thermal or inkjet printing with shop header and GST details.	Implemented
Payment Mode Filter	Filter sales records by payment method: Cash, UPI, Card, enabling quick reconciliation per payment channel.	Implemented
Status Filter	Filter invoices by status: Completed, Pending, Partially Paid, Returned for focused view of transaction states.	Implemented
Daily Summary Card	Aggregated daily sales metrics including total invoices, total revenue, and average order value displayed prominently.	Implemented
Weekly Summary Card	Weekly aggregated sales data with comparison to the previous week to identify trends and anomalies.	Implemented
Monthly Summary Card	Monthly performance overview with revenue, transaction count, and growth percentage against the prior month.	Implemented
WhatsApp Invoice Sharing	One-click sharing of invoice details to customer via WhatsApp using pre-formatted message templates.	Implemented
GST Calculation	Automatic CGST and SGST computation at configurable rates (default 9% each) applied to line items and invoice totals.	Implemented
Product Search in POS	Real-time product search by name, SKU, or barcode during invoice creation for rapid item lookup.	Implemented

4. Inventory (1,745 lines)

Feature	Description	Status
Product CRUD	Complete create, read, update, and delete operations for inventory products with fields for name, SKU, category, price, and stock quantity.	Implemented
Quick Stock Adjust Dialog	Rapid stock quantity adjustment via a lightweight dialog without needing to open the full product edit form.	Implemented
CSV Import	Bulk import products from a CSV file with field mapping, validation, and error reporting for failed rows.	Implemented
Bulk Import	Dedicated bulk import interface for adding multiple products at once with batch processing and progress indication.	Implemented
Low Stock Alerts	Configurable low-stock threshold per product with visual alert badges and a dedicated low-stock filter view.	Implemented
Frame Size Display	Display and manage frame size attributes (width, bridge, temple length) for optical frame products.	Implemented
Search and Filter	Multi-field search across product name, SKU, and category with combined filter support for refined results.	Implemented
Pagination	Configurable pagination for the product list with options for 10, 25, 50, or 100 items per page.	Implemented
Category Management	Organize products into categories (Frames, Lenses, Sunglasses, Accessories, Contact Lenses) for structured browsing.	Implemented
Price and Cost Tracking	Track both selling price and cost price per product for margin and profitability analysis.	Implemented

5. Prescriptions

Feature	Description	Status
Customer Prescription Management	Create, view, edit, and delete prescriptions linked to specific customer records for complete eyecare documentation.	Implemented
Spherical/Cylindrical Values	Record SPH and CYL values for both left (OS) and right (OD) eyes with decimal precision and positive/negative indicators.	Implemented
Axis Recording	Capture the axis degree value (0-180) for cylindrical correction in each eye as part of the full prescription data.	Implemented
PD Measurement	Record Pupillary Distance measurements for accurate lens fitting, with support for separate near and distance PD values.	Implemented
Lens Type Tracking	Track the prescribed lens type including Single Vision, Bifocal, Progressive, and Reading lenses for each prescription.	Implemented
Prescription Date and Doctor	Record the date of examination and the prescribing optometrist/doctor name for reference and follow-up.	Implemented
Add Power for Bifocal/Progressive	Additional near vision add power field for bifocal and progressive lens prescriptions.	Implemented
Prescription History	View complete historical list of all prescriptions for a customer with date-based sorting and comparison capability.	Implemented

6. Lab Orders (1,102 lines)

Feature	Description	Status
Lab Order Creation	Create new lab orders linked to a customer and prescription with lens specifications and frame selection.	Implemented
Status Tracking	Track each lab order through its lifecycle: Pending, In Progress, Ready, and Delivered with timestamp recording.	Implemented
Order Search	Search lab orders by customer name, order ID, or prescription reference for quick retrieval.	Implemented
Due Date Tracking	Set and monitor expected completion dates for each lab order with overdue alerts for delayed orders.	Implemented
Lab Order Detail View	Comprehensive detail view showing prescription details, lens specs, frame info, and status history.	Implemented
Status Update Workflow	Simple status transition interface to move orders through the pipeline with automatic timestamp logging.	Implemented
Customer Notification	Notify customers via WhatsApp or SMS when their lab order status changes to Ready or Delivered.	Implemented

7. Appointments (1,593 lines)

Feature	Description	Status
Calendar Month View	Full month calendar grid showing all scheduled appointments with color-coded indicators for different appointment types.	Implemented
Calendar Week View	Detailed week view with hourly time slots for precise scheduling and conflict detection.	Implemented
Walk-in vs Scheduled Toggle	Distinguish between walk-in and pre-scheduled appointments with a clear toggle during creation.	Implemented
Purpose Type Selection	Select from predefined appointment purposes: Eye Exam, Frame Selection, Lens Fitting, Delivery, Follow-up, Walk-in.	Implemented
SMS Reminders	Automated SMS reminder sent to customers before their scheduled appointment time.	Implemented
WhatsApp Reminders	Automated WhatsApp reminder message sent to customers prior to their appointment.	Implemented
Recurrence Support	Create recurring appointments with daily, weekly, or monthly frequency patterns for regular follow-up visits.	Implemented
Status Management	Manage appointment lifecycle: Scheduled, Confirmed, Completed, Cancelled, No-Show with visual status badges.	Implemented
Date Picker Navigation	Quick navigation to any date via an integrated date picker component for fast appointment lookup.	Implemented
Appointment Search	Search appointments by customer name, phone, or purpose to locate specific bookings quickly.	Implemented
Time Slot Selection	Select specific time slots when creating appointments with visual availability indicators.	Implemented
Customer Linking	Associate each appointment with an existing customer record or create a new customer inline during booking.	Implemented

8. Staff (1,594 lines)

Feature	Description	Status
Full CRUD Operations	Create, read, update, and delete staff member records with personal details, role assignment, and contact information.	Implemented
Clock In/Out Attendance	Digital clock-in and clock-out functionality with live timer showing current shift duration.	Implemented
Live Timer Display	Real-time running timer on the staff dashboard showing elapsed time since clock-in for currently active staff.	Implemented
Commission Tracking	Automatic commission calculation per sale for commission-eligible staff roles with cumulative period totals.	Implemented
Salary Management	Generate salary records for any period with automatic calculation of base pay, commission, and deductions.	Implemented
Mark Salary as Paid	One-click action to mark generated salary records as paid with date stamping for payroll records.	Implemented
Role-Based Access Matrix	Five predefined roles: Owner, Admin, Optometrist, Sales Staff, and Assistant with granular permission controls per module.	Implemented
Performance Reports	Individual and team performance dashboards showing sales attributed, appointments handled, and revenue generated.	Implemented
Bank Details	Secure storage of staff bank account information for salary disbursement and record-keeping.	Implemented
Staff Attendance History	Complete attendance log with clock-in/out times, total hours, and date-range filtering for each staff member.	Implemented
Staff Directory	Searchable staff directory with role-based filtering and quick access to individual staff profiles.	Implemented

9. Campaigns (1,518 lines)

Feature	Description	Status
SMS Campaigns	Create and send bulk SMS campaigns to targeted customer segments with delivery tracking.	Implemented
WhatsApp Campaigns	Launch WhatsApp marketing campaigns using pre-approved message templates for compliance.	Implemented
Print Campaigns	Plan and track print media campaigns including flyers, banners, and newspaper advertisements.	Implemented
Online Campaigns	Manage digital marketing campaigns across online channels with cost and reach tracking.	Implemented
WhatsApp Templates (4 Pre-built)	Four pre-configured WhatsApp message templates: Promotional Offer, Birthday Greeting, Due Reminder, Collection Ready Notification.	Implemented
SMS Character Tracker	Real-time character count with 160-character per SMS segment limit indication and segment count display.	Implemented
ROI Tracking	Track return on investment per campaign with cost input, revenue attribution, and profit/loss calculation.	Implemented
Customer Segment Targeting	Target campaigns to specific segments: All Customers, Regular, Wholesale, New, or High Value customers.	Implemented
Analytics Tab	Dedicated analytics view with performance bar charts showing delivery rates, open rates, and conversion metrics.	Implemented
Birthday Greetings	Automated birthday greeting campaigns that trigger based on customer birthday data.	Implemented
Due Reminders	Automated payment due reminder campaigns targeting customers with outstanding balances.	Implemented
Collection Ready Notifications	Automated notifications sent to customers when their lab orders or prescriptions are ready for collection.	Implemented
Mobile-Responsive Card View	Campaign cards displayed in a responsive grid layout optimized for both mobile and desktop viewing.	Implemented
Campaign Status Management	Track campaigns through Draft, Scheduled, Active, Completed, and Cancelled states.	Implemented

10. Accounting (2,594 lines)

Feature	Description	Status
GST Summary (CGST/SGST/IGST)	Comprehensive GST breakdown showing collected CGST, SGST, and IGST totals with period filtering and export capability.	Implemented
Expense Tracking	Record and categorize business expenses with date, amount, category, and description fields for complete financial tracking.	Implemented
Dues Management	Track customer outstanding balances with aging analysis showing current, 30-day, 60-day, and 90+ day overdue amounts.	Implemented
Profit/Loss Calculation	Automatic profit and loss computation from revenue and expense data with configurable reporting periods.	Implemented
Payment Tracking	Record and track incoming and outgoing payments with reference to sales invoices, purchase orders, and expense entries.	Implemented
Daily Cash Summary	End-of-day cash position summary showing opening balance, cash inflows, cash outflows, and closing balance.	Implemented
Expense Categories	Predefined expense categories (Rent, Utilities, Salaries, Supplies, Marketing, Maintenance, Other) with custom category support.	Implemented
Financial Date Filters	Filter all accounting data by custom date ranges, or use preset periods: Today, This Week, This Month, This Quarter, This Year.	Implemented
Accounting Dashboard	Central financial overview with key metrics: total revenue, total expenses, net profit, pending dues, and GST liability.	Implemented
Payment Method Breakdown	Revenue breakdown by payment method (Cash, UPI, Card) for bank reconciliation and cash flow management.	Implemented

11. Reports (2,359 lines)

Feature	Description	Status
Sales Trend Report	Visual and tabular representation of sales trends over customizable time periods with daily, weekly, and monthly aggregation options.	Implemented
Top Products Report	Ranked listing of products by sales quantity and revenue with period filtering and comparison capabilities.	Implemented
Top Customers Report	Customer ranking by total purchase value, purchase frequency, and average order value for loyalty program insights.	Implemented
Inventory Turnover Report	Analysis of how quickly inventory items are sold and replaced with turnover ratio calculations per product.	Implemented
Revenue Comparison	Side-by-side revenue comparison between two periods (e.g., this month vs. last month) with variance percentages.	Implemented
Product Performance Report	Individual product performance metrics including units sold, revenue generated, profit margin, and stock status.	Implemented
Customer Acquisition Report	Track new customer additions over time with source attribution and segment distribution analysis.	Implemented
Date Range Selection	Flexible date range picker for all reports with preset options for quick access to common time periods.	Implemented
Report Export	Export any generated report to CSV format for further analysis in external spreadsheet applications.	Implemented
Summary Statistics	Aggregate statistics (totals, averages, min, max) displayed prominently at the top of each report for quick insights.	Implemented

12. Purchase Orders (763 lines)

Feature	Description	Status
PO Creation	Create purchase orders with supplier selection, product line items, quantities, unit prices, and expected delivery dates.	Implemented
Supplier Linking	Associate each purchase order with a supplier from the supplier directory for organized procurement tracking.	Implemented
Status Tracking	Track purchase order status through its lifecycle: Draft, Sent, Confirmed, Partially Received, Received, Cancelled.	Implemented
PO Search and Filter	Search purchase orders by supplier name, PO number, or product name with status-based filtering.	Implemented
Order Detail View	Comprehensive detail view showing all line items, supplier information, status history, and received quantities.	Implemented
Inventory Update on Receipt	Automatic inventory stock increment when a purchase order is marked as received.	Implemented
PO Print View	Print-optimized purchase order layout suitable for sending to suppliers or maintaining physical records.	Implemented

Technical Architecture

API Routes (21 Endpoints)

Method	Route	Description
GET	/api/customers	List all customers with pagination and filters
POST	/api/customers	Create a new customer record
GET	/api/customers/[id]	Get customer by ID with related data
PUT	/api/customers/[id]	Update customer record
DELETE	/api/customers/[id]	Delete customer record
GET	/api/sales	List all sales/invoices with filters
POST	/api/sales	Create a new sale/invoice
GET	/api/sales/[id]	Get sale details by ID
PUT	/api/sales/[id]	Update sale record (status, return)
GET	/api/inventory	List all inventory products
POST	/api/inventory	Add a new product to inventory
PUT	/api/inventory/[id]	Update product details and stock
DELETE	/api/inventory/[id]	Remove product from inventory
GET	/api/appointments	List appointments with date filters
POST	/api/appointments	Create a new appointment
PUT	/api/appointments/[id]	Update appointment details/status
GET	/api/staff	List all staff members
POST	/api/staff	Add a new staff member
PUT	/api/staff/[id]	Update staff record
GET	/api/campaigns	List campaigns with status filter
POST	/api/campaigns	Create and launch a new campaign

Prisma Models (16 Models)

Model	Key Fields
Customer	name, phone, email, address, group, birthday, notes
Sale	invoiceNumber, customerId, subtotal, tax, total, status, paymentMode
SaleItem	saleId, productId, quantity, price, discount
Product	name, sku, category, price, costPrice, stock, minStock
Prescription	customerId, sphR, cylR, axisR, sphL, cylL, axisL, pd, addPower, lensType
LabOrder	customerId, prescriptionId, status, dueDate, notes
Appointment	customerId, date, time, purpose, type, status, recurrence
Staff	name, phone, email, role, salary, bankDetails, status
Attendance	staffId, clockIn, clockOut, totalHours, date
Commission	staffId, saleId, amount, date
Salary	staffId, basePay, commission, deductions, total, status, period
Campaign	name, type, status, segment, message, scheduledAt, sentAt
Expense	category, amount, description, date, reference
PurchaseOrder	supplierId, status, total, expectedDate, notes
PurchaseOrderItem	purchaseOrderId, productId, quantity, unitPrice
Supplier	name, phone, email, address, notes

Keyboard Shortcuts and UX Features

Ctrl+K: Global search - opens the command palette for quick navigation and search across all modules.

1 through 0: Section switching - number keys 1 through 0 switch between the ten primary CRM sections instantly.

The application supports dark mode with a system-preference-aware toggle that persists the user's choice. Page transitions are handled with smooth CSS animations to provide a polished, native-app-like feel during navigation between sections.

Mobile and UX Features

The Lotus Vision Opticals CRM is built with a mobile-first approach, ensuring that every feature is fully functional and comfortably usable on smartphones, tablets, and desktop browsers. Below are the key mobile and user experience design decisions implemented across the application.

Feature	Description	Status
44px Touch Targets	All interactive elements including buttons, links, form inputs, and tab controls maintain a minimum touch target size of 44x44 pixels, meeting WCAG accessibility guidelines and ensuring comfortable use on touch devices.	Implemented
Mobile-Responsive Card Layouts	Staff directory, campaign list, and sales summary screens use responsive card-based layouts that adapt from a single-column view on mobile to multi-column grids on larger screens.	Implemented
Horizontal Scroll Tables	All wide data tables (customers, sales, inventory, reports) feature horizontal scroll containers on narrow viewports, ensuring no data columns are hidden or truncated.	Implemented
Collapsible Sidebar with Mobile Drawer	The main navigation sidebar collapses to a hamburger-triggered drawer on mobile devices, providing full access to all sections without consuming screen real estate.	Implemented
Dark Mode Support	A system-aware dark mode toggle allows users to switch between light and dark themes. The preference is persisted in local storage and applied consistently across all views.	Implemented
Page Transitions	Smooth CSS-based page transition animations provide visual continuity during navigation, creating a polished single-page application experience.	Implemented